

TSX Trading Advisor

Technical Reference Manual

Architecture, Configuration, Data Flow, AI Logic and Trading Rules

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Mode: SHORT TERM (1-Week Momentum)

This document is auto-generated and reflects the live codebase. It is intended for the system operator only.

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1. System Overview

TSX Trading Advisor is a zero-cost, single-user autonomous stock trading assistant for Canadian equities on the Toronto Stock Exchange. It continuously monitors a user-defined portfolio, fetches live market data and macro news, and delivers scheduled AI-generated BUY/SELL/HOLD recommendations and End-of-Day recaps via Telegram.

The system runs entirely locally on a Windows machine using Python (Miniforge). There are no cloud services, no subscription fees, and no external AI API keys. All AI reasoning is performed natively by the Antigravity agent session.

Key Characteristics

- Horizon: SHORT TERM (1-week momentum and mean-reversion setups)
- Data: Live prices, indicators, and macro context fetched at 30-minute intervals
- Delivery: Telegram Bot notifications directly to the operator's phone
- Control: Operator logs trades via natural language text messages to the bot
- Portfolio: Configurable list of TSX tickers and commodity monitors in config.json
- Scheduling: 30-minute scans (9:30 AM - 4:00 PM ET) + EOD summary (4:05 PM ET)

What the System Does NOT Do

- It does NOT execute trades automatically. All trades must be placed manually by the operator.
- It does NOT access broker APIs or hold private brokerage credentials.

(i) **IMPORTANT:** This tool provides AI-generated trading suggestions only. All investment decisions are the sole responsibility of the operator.

2. Architecture and Data Flow

2.1 High-Level Architecture

The system is split into three distinct layers: a deterministic Data Engine, an AI Reasoning Layer, and a Telegram Delivery Layer. Communication is mediated via local JSONL queue files and standard output streams, creating a clean, fault-tolerant bridge.

```

[Telegram User]
|
v (sends message)
[telegram_daemon.py] <--- One-Shot Mode (streams NEW_MESSAGE to stdout)
|
+-----+ (wakes agent session directly or via queue)
v               v
[Antigravity Agent] <--- The AI brain
|
+--> runs send_interval_scan.py / market_analyzer.py
+--> reasons natively, evaluates rules & macro context
|
v (writes response)
[outgoing_queue.jsonl]
|
v (polled every 2s by)
[telegram_daemon.py] --> [Telegram User]
                    +--> log_advice(advice_history.txt)

```

2.2 Scheduled Analysis Flow

Task	Cron Expression	Description
30-min Market Scan	* / 30 9-16 * * 1-5	Full portfolio scan, true P&L, BUY/SELL signals
End-of-Day Summary	5 16 * * 1-5	Portfolio recap, realized trades, next-day watchlist

2.3 Advice History and Memory

Every message sent through `outgoing_queue.jsonl` is automatically logged to `advice_history.txt` by `telegram_daemon.py`. This provides the AI with rolling memory of its own prior advice. The `get_cleaned_advice_history()` function trims the file on each dossier generation to retain: up to the last 5 entries from today plus the final entry from the prior 5 calendar days.

(i) An OS-level file lock (`advice.lock`) prevents corruption from concurrent writes. Stale lock files older than 30 seconds are automatically removed.

3. File Inventory

3.1 Core Python Scripts

File	Role
market_analyzer.py	Core data engine: prices, indicators, tripartite macro news, dossier generator
send_interval_scan.py	30-min interval scan script: calculates equity/P&L, checks signals, dispatches to Telegram
send_eod_summary.py	EOD summary script: cumulative realized P&L, trade recap, commodity wrap, watchlist
telegram_daemon.py	Telegram bot relay: one-shot listener, polls outgoing queue, sends messages, logs advice
file_watcher.py	Daemon monitor bridge: loops continuously, watches incoming messages to wake agent
generate_pdf.py	Documentation generator: produces this technical manual from source
queue_reply.py	Utility script: quickly queues responses into outgoing_queue.jsonl

3.2 Configuration and Data Files

File	Purpose
config.json	Portfolio ticker list, macro commodities (CL=F, GC=F, HG=F), mode (SHORT), stop-loss %
trades.csv	Trade ledger: BUY/SELL/DEPOSIT/DIVIDEND history, Adjusted Cost Base, running cash
advice_history.txt	Rolling log of all AI advice sent to Telegram
.env	Secrets: TELEGRAM_BOT_TOKEN, TELEGRAM_CHAT_ID, TELEGRAM_PIN,
.gitignore	Excludes .env, *.lock, *.log, __pycache__, cache JSONs, queue files from git
requirements.txt	Python dependencies for pip installation (yfinance, pandas, fpdf2, python-dotenv)
AGENT_RESTORE_INSTRUCTIONS.	Step-by-step AI session restore guide for agent restarts
.agents/AGENTS.md	Persistent agent rules: trading strategy overrides and security protocol

3.3 Runtime Queue Files (transient)

File	Purpose
incoming_queue.jsonl	Telegram messages from user, waiting for AI processing
incoming_processing.jsonl	Atomic rename during processing (prevents double-read)
outgoing_queue.jsonl	AI responses queued for Telegram delivery
outgoing_processing.jsonl	Atomic rename during daemon polling
advice.lock	OS-level mutex protecting advice_history.txt writes

(i) Queue files and cache JSON files are ephemeral and excluded by .gitignore.

4. Configuration Reference (config.json)

config.json provides a streamlined configuration structure:

```
{
  "mode": "SHORT",
  "portfolio": [
    "CNQ.TO",
    "ABX.TO",
    "CP.TO",
    "CLS.TO",
    "SHOP.TO",
    "CNR.TO"
  ],
  "macro_commodities": [
    "CL=F",
    "GC=F",
    "HG=F"
  ],
  "stop_loss_pct": 5.0
}
```

Fields

Field	Type	Description
mode	string	"SHORT" (1-week momentum) or "MEDIUM" (2-4 week swing). Controls
portfolio	array	Array of equity ticker symbols on the TSX to actively track and analyze.
macro_commodities	array	Key commodities (Crude Oil, Gold, Copper) tracked for sector sentiment.
stop_loss_pct	float	Baseline stop-loss threshold percentage (typically 5.0%).

Consolidated Tripartite Macro Briefing

Instead of spamming individual news feeds for dozens of stocks, the engine synthesizes a single high-level tripartite macroeconomic briefing across:

1. Canada Macro: Bank of Canada policy, TSX composite trends, domestic CPI/inflation.
2. USA Macro: Federal Reserve rate expectations, US labor/inflation data, S&P 500 sentiment.
3. World / Global Macro: Geopolitical events, global trade dynamics, central banks, and commodity supplies.

5. Market Dossier Structure

The dossier is a structured plain-text document generated by `market_analyzer.generate_dossier()` on every scheduled scan or manual analysis request. It contains five key sections.

Section 1: Mode and Timestamp

```
=== CURRENT MODE: SHORT TERM (1-WEEK HORIZON) ===
Date: 2026-10-05 15:30:00 ET
```

Section 2: Portfolio and Pricing (per ticker)

Field	Description
Shares Owned	From trades.csv ledger
Avg Purchase Price (ACB)	Adjusted Cost Base computed from trade history
Current Market Price	Live price with multi-tier fallback (1d -> 5d -> ACB)
Previous Close	Prior day closing price
Gap %	Overnight gap: (Open - PrevClose) / PrevClose
Open / High / Low	Intraday OHLC from yfinance
Intraday Trend %	How far current price is above the Low of Day
Unrealised P&L	Dollar and % profit/loss on current position
STOP-LOSS Alert	Injected if price is > stop_loss_pct% below ACB
Vol Ratio (10-day)	Volume divided by 10-day average volume
EMA Crossover	Explicit BULLISH / BEARISH / NEUTRAL label (EMA5 vs EMA9)
EMA (5-day)	5-day Exponential Moving Average
EMA (9-day)	9-day Exponential Moving Average
RSI (7-day)	7-period Relative Strength Index
ATR (5-day)	Average True Range (5-period)
Bollinger Lower/Mid/Upper	20-period Bollinger Bands (2 std deviations)

Section 3: Macro Commodities

Commodity symbols (CL=F, GC=F, HG=F) are displayed with current price and daily % change to align sector entries.

Section 4: Tripartite Macroeconomic News Briefing

- Structured into Canada, USA, and World news pillars.
- [NEW] (<=4h), [RECENT] (4-12h), [OLD] (12-48h) publication badges.
- Category badges: [MACRO], [CENTRAL_BANK], [COMMODITY], [TRADE].

Section 5: Previous Advice History

A rolling window of past AI analysis. The LLM receives up to the last 5 entries from today plus the final entry from the preceding 5 calendar days. Preserves cross-session continuity.

6. AI Trading Rules and Logic

Trading rules are defined across system prompt templates in market_analyzer.py and persistent overrides in .agents/AGENTS.md. AGENTS.md rules take strict precedence.

6.1 Core Antigravity Execution Guidelines

- Target Sell & Stop Loss on BUY

Every BUY must provide a Target Sell Price (Mid-BB or +1.5 ATR target, whichever is > purchase price) and -5% Stop Loss.

- Macro-Alignment Rule

Before buying commodity-linked stocks (AEM, CNQ), the underlying commodity (GC=F, CL=F) MUST be green on the day.

- Strict Overbought Filter

If 7-day RSI > 75, NEVER BUY. If RSI is 65-75, only buy if Vol Ratio > 1.25x.

- ADX Trend Filter

BUY requires 14-day ADX > 20 unless Vol Ratio > 1.25x (volume surge exception).

- Low-Volume Scaled Entry

Bullish EMA crossover with Vol Ratio ≤ 1.25x: allowed only with scaled position (25-50% allocation).

- Stagnation Exit

Position held >4 trading days, negative, and flashes bearish EMA crossover: SELL immediately to recycle capital.

- Profit Target Breakeven Flag

If an open position gains >2.0% above purchase price, flag to user to raise Stop Loss to Breakeven.

- Time-of-Day Filter

Never initiate new BUY signals between 11:30 AM and 2:00 PM ET (lunch fake-out zone).

- Volume Capitulation

Vol Ratio > 1.25x during a drop indicates panic selling exhausted; valid reversal signal.

- Intraday Reversal Trigger

Intraday drop >1.25% below open + RSI < 35 with immediate 2-interval rebound: allows speculative entry.

7. Trading Prompt System Instructions

The AI formats responses concisely for mobile Telegram consumption, with dedicated scan and summary structures.

7.1 30-Minute Interval Scan Format

- Unrealized P&L: Single summary line total at the top of the scan.
- Portfolio Status: Quick bullet for each open position showing ACB, current price, and gain/loss.
- Actionable Signals: BUY/SELL alerts with precise Target Price and Stop Loss calculations.
- Position Sizing: Explicit labeling of Low-Volume Scaled Entries (25-50% allocation).

7.2 End-of-Day (EOD) Summary Format

- Total Unrealized P&L and Portfolio Equity.
- Today's Realized Activity: Profit/loss from any closed trades.
- Cumulative Performance: Active guidance realized P&L, trade count, and win rate.
- Commodity & Macro Wrap: Gold, Crude Oil, and copper daily status.
- Next-Day Watchlist: Actionable setups identified for the following morning's open.

8. Telegram Security Protocol

Because the Telegram bot relays messages directly to the AI agent which has filesystem and command capabilities, a cryptographic PIN authentication protocol prevents remote code execution.

8.1 Remote Modification Challenges

- Editing any Python (.py) source file.
- Modifying .agents/AGENTS.md, config.json, or .env.
- Running arbitrary shell, system, or PowerShell commands.
- EXCLUSION: Appending or updating trades.csv for user trade logging is exempt.

8.2 Challenge & Verification Workflow

```

[Telegram User asks for code/system change]
    |
    v
AI halts & writes PIN Challenge to Telegram
(includes TELEGRAM_PIN_HINT from .env)
    |
    v
[User replies with 8-digit PIN]
    |
    v
AI verifies against TELEGRAM_PIN in .env
Match: Executes change | Mismatch: Rejects request
  
```

8.3 Local User Exception

Requests originating directly from the local Antigravity chat interface are fully authenticated and exempt from PIN challenges.

8.4 Environment Secrets (.env)

Variable	Description
TELEGRAM_BOT_TOKEN	Bot authentication token issued by @BotFather.
TELEGRAM_CHAT_ID	Authorized Telegram user ID. All unauthorized IDs are rejected.
TELEGRAM_PIN	8-digit secret PIN required for remote code changes.
TELEGRAM_PIN_HINT	Hint phrase displayed when requesting the PIN challenge.
(i) NEVER commit .env to git. Always verify .gitignore contains .env.	

9. Session Restore Procedure

When restarting or restoring an agent session, follow this streamlined procedure:

Step 1: Start Telegram Daemon

Run in the background via run_command:

```
C:\Users\Aamir\miniforge3\python.exe telegram_daemon.py
```

Step 2: Start File Watcher

Ensure file_watcher.py is active as a daemon process:

```
C:\Users\Aamir\miniforge3\python.exe file_watcher.py
```

Step 3: Setup Automated Cron Schedules

Task	Cron Expression	Script Executed
30-min Market Scan	* /30 9-16 * * 1-5	send_interval_scan.py
End-of-Day Summary	5 16 * * 1-5	send_eod_summary.py

Step 4: Operational Health Check

- Run `manage_task(Action='list')` to verify daemons and schedules are running.
- Send `/status` via Telegram to verify daemon connectivity and queue readiness.

10. Known Limitations and Operational Notes

10.1 Market Data & API Fallbacks

- yfinance TSX quotes have ~15-minute standard market delay.
- Empty 1-day history frames at open are protected by automated 5-day and cost-basis fallbacks.
- Intraday High/Low/Open reflect the current day's active bar.

10.2 Trade Ledger Integrity (trades.csv)

- All trades are logged in chronological order.
- Cash balance is derived from the latest row's Cash_Balance column.
- Adjusted Cost Base (ACB) is rolled forward cumulatively across all buy/sell transactions.

10.3 Background Daemons

- file_watcher.py runs continuously to bridge queue changes.
- telegram_daemon.py processes incoming user instructions and flushes outgoing responses.
- Long Telegram messages are automatically chunked at 4,000 characters to comply with API limits.

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